



UNIVERSITI TUNKU ABDUL RAHMAN

LEE KONG CHIAN FACULTY OF ENGINEERING AND SCIENCE

UECS1044/1144 OBJECT-ORIENTED APPLICATION DEVELOPMENT

Group Assignment - Library E-Book Lending & Reservation System

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Introduction

System

Library E-Book Lending & Reservation System

Purpose

This document is intended to define the functional requirements of the Library E-Book Lending & Reservation System, a console-based Java application. The system provides user reservation and reading of e-books in a virtual library with an administration facility for managing users, books, and reservations. Each feature of the system is explained in detail here as an implementation, testing, and future development reference guide.

Scope

This system imitates an e-book lending website in electronic form to be utilized for educational or public use. It supports:

- User registration, authentication, and role-based access control
- Display and management of digital books and user accounts
- Reservation and reading of digital books
- Automatic and manual reservation status update
- Administrative control over library catalog and user management

The software runs on the console and uses text files for simulating permanent storage (users.txt, books.txt, reservations.txt, etc.). The software is to use advanced object-oriented programming concepts like inheritance, polymorphism, interfaces, file operations, and exception handling.

This document has:

- Functional Requirements
- Input/Output specifications
- Preconditions and user actions
- Status transition logic

No GUI design, networked functionality, or integration with external systems such as email or real-time notifications.

Target Audience

This document is for:

- Team members implementing system design and programming
- Professors marking the design and correctness of the assignment

Assumptions

- The system will be implemented in a controlled environment.
- All users interact with the system through the command-line interface.
- Storage using files is utilized instead of a full database.
- Dates are placed in a standard and valid format (e.g., YYYY-MM-DD).
- Each e-book is delivered as a .txt file in a specified books/ directory.
- Only a single user can access the system at any given time.
- System time is depended upon and used to automatically update reservation status.

System Requirements Specification (SRS)

System Features

Feature	Purpose	Tasks (Functional Requirements)
Authentication	Manage secure user access	FR02 Sign up FR03 Log in FR09 Logout
User Profile Management	Allow users and admins to manage user credentials and roles	FR04 View user details FR05 Update username FR06 Update password FR07 Update role FR08 Delete user
User Management (Admin)	Admin-level management of user data	FR34 View users with pagination FR35 Sort users FR36 Filter users FR37 Select user FR38 Create user
System Initialization	Ensure data consistency at startup through automatic updates	FR01 Automatically update reservation statuses
Reservation Overview	Display and manage reservations list with filters, sorting, and pagination	FR10 View reservations with pagination FR11 Sort reservations FR12 Filter reservations FR13 Select reservation
Reservation Details	View and act on a specific reservation	FR14 View reservation FR15 View book FR16 Update status FR17 Update start date

		FR18 Update end date FR19 Delete reservation
Books Overview	Browse available books in the system	FR20 View books with pagination FR21 Sort books FR22 Filter books FR23 Select book
Book Management	Manage book records (Admins only)	FR24 Add book FR28 Update title FR29 Update author FR30 Update genre FR31 Update release date FR32 Update filename FR33 Delete book
Book Details	View, reserve, or read an individual book	FR25 View book FR26 Reserve book FR27 Read book

Functional Requirements

Authentication Pages

FR01 Automatically update statuses of reservations

Description: Upon starting the system, update the reservations statuses

1. From Approved to Active if the current date is larger than or equal to the start date
2. From Active to Expired if the current date is larger than the end date
3. From Pending to Denied if the current date is larger than the start date
4. the current status otherwise

Input: None

Expected Output: Updated reservations.txt

Precondition: None

User interaction: Starting the system

FR02 Sign up

Description: Register a new user

Inputs: username; password; confirm password

Expected Output: Confirmation message, new user added to users.txt

Precondition: username must be unique; passwords must match

User interaction: User enters the inputs via console prompts

FR03 Log in

Description: Authenticate a user

Input: username; password

Expected Output: Confirmation message; Direct user to home page

Precondition: username must exist; username and password must match

User interaction: Users enter the inputs via console prompts. Users will be notified to contact the admins if they forgot their password.

My Account/User Pages

FR04 View user details

Description: show username and role

Input: username

Expected output: username, role

Precondition: username must exist

User interaction: Console output

FR05 Update username

Description: members can update their username only while admins can update all users' username.

Input: new username

Expected output: Confirmation message; updated users.txt

Precondition: username must be unique.

User interaction: User enters the inputs via console prompts.

FR06 Update password

Description: members can update their password only while admins can update all users' password.

Input: username, old password, new password, confirm new password

Expected output: Confirmation message; updated users.txt

Precondition: old password must be correct, new passwords must match.

User interaction: User enters the inputs via console prompts

FR07 Update role

Description: admins can update any user's role

Input: username, role, confirmation to switch role from admin to user or vice versa

Expected output: Confirmation message; updated users.txt

Precondition: the user is an admin

User interaction: User enters the inputs via console prompts

FR08 Delete user

Description: members can delete their account only while admins can delete any user's account

Input: username

Expected output: Confirmation message; update users.txt; log out the user if the deleted user is the current user

Precondition: username must exist

User interaction: User enters the username via console prompt to confirm delete.

FR09 Logout

Description: log out the current user

Input: None

Expected output: Confirmation message, logout the user

Precondition: the user must be logged in

User interaction: User selects log out via console prompt

Reservations Page

FR10 View reservations with pagination

Description: Display the top 10 reservations by descending reservation date for the user if the user is a member or all users if the user is an admin, or nothing if page number exceeds the number of pages

Input: Page number (default: 1)

Expected output: A list of reservations summaries for the page

Precondition: None

User interaction: User enters a page number via console prompt

FR11 Sort reservations

Description: Sort the filtered reservations by id, book title, username, reservation date, status, start date or end date

Input: sort field, sort order

Expected output: sorted reservation list with pagination

Precondition: None

User interaction: User selects the inputs via console prompts

FR12 Filter reservations

Description: Filter reservations by

- Partial match: book title, username
- Exact match: id, status
- Date range: reservation date, start date or end date

Input: filter field, filter input

Expected output: filtered reservation list with pagination

Precondition: None

User interaction: User enters/selects the inputs via console prompts

FR13 Select reservation

Description: Select a reservation to view or modify, where members can only select their reservation while admins can select any reservation.

Input: reservation id

Expected output: direct the user to Reservation page with the reservation id

Precondition: Reservation id must exist

User interaction: User enters the reservation id via console prompts

Reservation Page

FR14 View reservation

Description: View the reservation details

Input: reservation id

Expected output: reservation details

Precondition: Reservation id must exist

User interaction: console displays the reservation details

FR15 View book

Description: View the book of the reservation.

Input: book id

Expected output: view the book details of the reservation

Precondition: book id must exist

User interaction: User select View Book via console prompts

FR16 Update status

Description:

1. Admins can update any reservation's status to Approved or Denied if the current status is Pending
2. Members can update only their reservation's status to
 - a. Cancelled if the current status is Pending
 - b. Returned if the current status is Approved or Active

Input: reservation id, new status

Expected output: Confirmation message. Updated reservations.txt

Precondition: reservation id must exist

User interaction: User enters inputs via console prompts

FR17 Update start date

Description:

1. Admins can update any reservation's start date
2. Members can update their reservation's start date only

Input: reservation id, status, new start date

Expected output: Confirmation message. Updated reservations.txt

Precondition: reservation id must exist, status must be Pending, new start date must be after the reservation date and before the end date

User interaction: User enters inputs via console prompts

FR18 Update end date

Description:

1. Admins can update any reservation's end date
2. Members can update their reservation's end date only

Input: reservation id, status, new end date

Expected output: Confirmation message. Updated reservations.txt

Precondition: reservation id must exist, status must be Pending, new end date must be after the start date

User interaction: User enters inputs via console prompts

FR19 Delete reservation

Description: Only admins can delete reservation

Input: reservation id

Expected output: Confirmation message. Updated reservations.txt

Precondition: reservation id must exist, the user is an admin

User interaction: User re-enters the reservation id via console prompts to confirm reservation

Books page

FR20 View books with pagination

Description: Display the top 10 books by alphabetical order

Input: Page number (default: 1)

Expected output: A list of book summaries for the page

Precondition: None

User interaction: User enters a page number via console prompt

FR21 Sort books

Description: Sort the filtered books by id, title, author, genre, released date

Input: sort field, sort order

Expected output: sorted book list with pagination

Precondition: None

User interaction: User selects the inputs via console prompts

FR22 Filter books

Description: Filter books by

- Partial match: title, author, genre
- Exact match: id
- Date range: released date

Input: filter field, filter input

Expected output: filtered book list with pagination

Precondition: None

User interaction: User enters the inputs via console prompts

FR23 Select book

Description: Select a book to view more details

Input: book id

Expected output: direct the user to Book page with the book id

Precondition: book id must exist

User interaction: User enters the book id via console prompts

FR24 Add book

Description: Add a new book

Inputs: title, author, genre, released date, filename

Expected Output: Confirmation message, new book added to books.txt

Precondition: .\books\[filename].txt must exist

User interaction: User enters the inputs via console prompts

Book page

FR25 View book

Description: View the book details

Input: book id

Expected output: book details

Precondition: book id must exist

User interaction: console displays the book details

FR26 Reserve book

Description: Reserve the book for a user

Input: book id, username, reservation date, start date, end date

Expected output: Confirmation message, updated reservations.txt

Precondition: book_id, username exists, end date >= start date >= reservation date

User interaction: User enters the inputs via console prompts

FR27 Read book

Description: Display the book by pages, where each page has 20 lines

Input: filename, page number (default: 1)

Expected output: the book's page

Precondition: the user has reserved the book or the user is an admin

User interaction: User enter the page number via console prompts

FR28 Update title

Description: Update a book's title

Input: book id, new title

Expected output: confirmation message, updated books.txt

Precondition: the user is an admin

User interaction: User enters the inputs via console prompts

FR29 Update author

Description: Update a book's author

Input: book id, new author

Expected output: confirmation message, updated books.txt

Precondition: the user is an admin

User interaction: User enters the inputs via console prompts

FR30 Update genre

Description: Update a book's genre

Input: book id, new genre

Expected output: confirmation message, updated books.txt

Precondition: the user is an admin

User interaction: User enters the inputs via console prompts

FR31 Update release date

Description: Update a book's release date

Input: book id, new release date

Expected output: confirmation message, updated books.txt

Precondition: the user is an admin

User interaction: User enters the inputs via console prompts

FR32 Update filename

Description: Update a book's filename

Input: book id, new filename

Expected output: confirmation message, updated books.txt

Precondition: the user is an admin, .\books\[filename].txt must exist

User interaction: User enters the inputs via console prompts

FR33 Delete book

Description: Delete a book

Input: book id

Expected output: confirmation message, updated books.txt

Precondition: the user is an admin

User interaction: User re-enters the book's id to confirm deletion via console prompt

Users page

FR34 View users with pagination

Description: Display the top 10 users by alphabetical order

Input: Page number (default: 1)

Expected output: A list of users for the page

Precondition: The user is an admin

User interaction: User enters a page number via console prompt

FR35 Sort users

Description: Sort the filtered users by username or role

Input: sort field, sort order

Expected output: sorted user list with pagination

Precondition: The user is an admin

User interaction: User selects the inputs via console prompts

FR36 Filter user

Description: Filter users by

- Partial match: username
- Exact match: role

Input: filter field, filter input

Expected output: filtered user list with pagination

Precondition: The user is an admin

User interaction: User enters/selects the inputs via console prompts

FR37 Select user

Description: Select a user to view more details

Input: username

Expected output: direct the user to User page with the username

Precondition: The user is an admin, username must exist

User interaction: User enters the username via console prompts

FR38 Create user

Description: Register a new user

Inputs: username; password; confirm password, role

Expected Output: Confirmation message, new user added to users.txt

Precondition: the user is an admin, username must be unique, passwords must match

User interaction: User enters the inputs via console prompts

Reservation Status Transition Table

From	To	Who Can Do It
Pending	Approved / Denied	Admin
Pending	Cancelled	Member
Approved	Returned	Member
Active	Returned	Member
Active	Expired (auto)	System
Approved	Active (auto)	System
Pending	Denied (auto)	System

UML Class Diagram

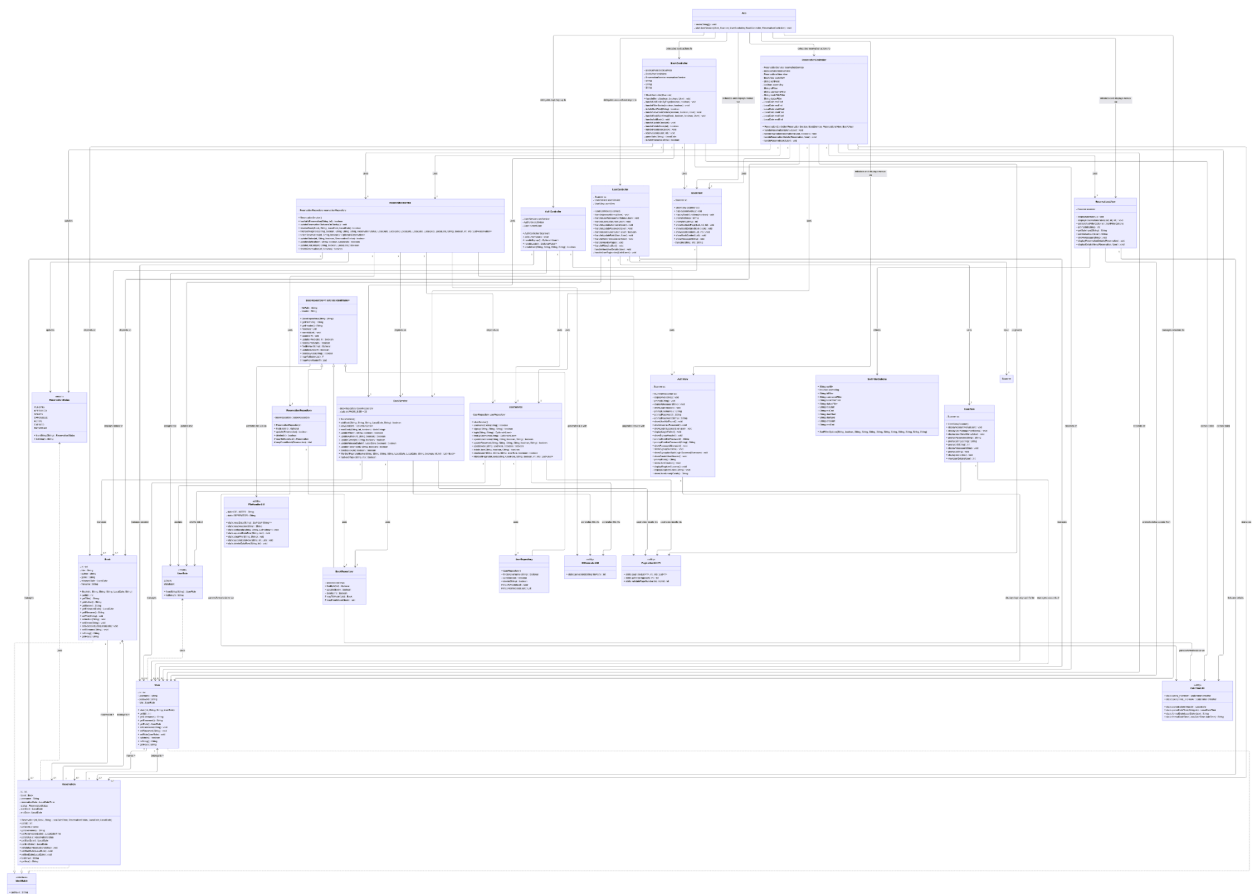
Tool name: Mermaid Chart

Source:

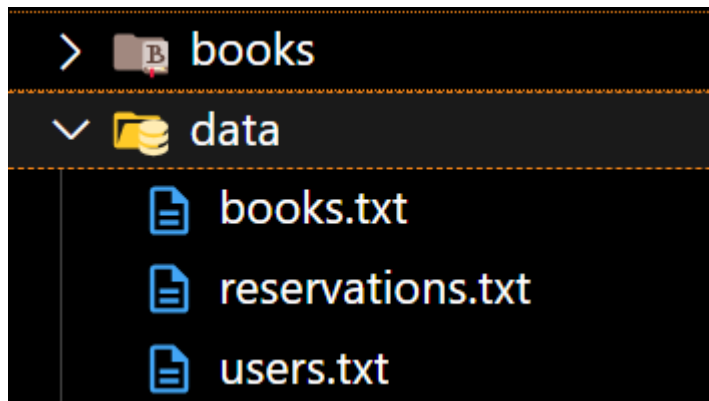
<https://www.mermaidchart.com/app/projects/16686229-19ff-4c35-99af-8f0c8608270f/diagrams/03fc0681-09e1-454c-ae1-c4bfb7331cf9/version/v0.1/edit>

NOTE:

1. For the most accurate, readable, and clear diagram, please copy the uml.mmd code under the uml folder of the project folder and paste it into a new project in [mermaidchart.com](https://www.mermaidchart.com).
2. The UML Class Diagrams are also available in the file format of .png and .svg.
3. The diagram below lacks readability due to the size of the diagram and is for reference only.
4. Please kindly contact the group leader if none of the uml class diagrams is readable.



Sample Input



Included in the source code's data or books folder.

books folder for the list of books content

books.txt for the list of books details

reservations.txt for the list of reservations

users.txt for the list of users

Sample Output

System Initialization

FR01 Automatically update statuses of reservations

```
20|76|js|2025-09-01T16:32|APPROVED|2025-09-02|2025-10-02
21|76|js|2025-09-01T16:32|ACTIVE|2025-09-02|2025-09-05
22|76|js|2025-09-01T16:32|PENDING|2025-09-02|2025-09-10
23|76|js|2025-09-01T16:32|ACTIVE|2025-09-02|2025-09-10
24|76|js|2025-09-01T16:32|PENDING|2025-10-02|2025-10-10

PS C:\cas\oad\oad> & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.15.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\cas\oad\oad\bin' 'App'
Welcome to the Library E-Book Lending & Reservation System!

=== Welcome ===
1. Login
2. Register
3. Exit
Enter your choice:
```

(Date of Testing: 2025-09-06) Status automatically updated

1. From APPROVED to ACTIVE when the current date is later than the start date
2. From ACTIVE to EXPIRED when the current date is later than the end date
3. From PENDING to DENIED when the current date is later than the start date
4. Otherwise

Authentication Pages

FR02 Sign up

```
Welcome to the Library E-Book Lending & Reservation System!

=== Welcome ===
1. Login
2. Register
3. Exit
Enter your choice: 2

=== Sign Up ===
Enter username: James
Enter password: james123
Confirm password: james123
Sign-up successful!
Auto-login successful!

Welcome, James!

[Logged in as: James | Role: member]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Logout
Enter your choice: █
```

Member sign up successfully.

```
Welcome to the Library E-Book Lending & Reservation System!
```

```
=== Welcome ===
```

- 1. Login
- 2. Register
- 3. Exit

```
Enter your choice: 2
```

```
=== Sign Up ===
```

```
Enter username: James
```

```
Enter password: james123
```

```
Confirm password: james12
```

```
Password and confirmation do not match.
```

```
=== Welcome ===
```

- 1. Login
- 2. Register
- 3. Exit

```
Enter your choice: █
```

Sign up with the wrong confirmation password - Sign up failed.

```
Welcome to the Library E-Book Lending & Reservation System!
```

```
=== Welcome ===
```

- 1. Login
- 2. Register
- 3. Exit

```
Enter your choice: 2
```

```
=== Sign Up ===
```

```
Enter username: James
```

```
Enter password: james22
```

```
Confirm password: james22
```

```
Registration failed: User already exists.
```

```
=== Welcome ===
```

- 1. Login
- 2. Register
- 3. Exit

```
Enter your choice: █
```

Sign up with existing username - Sign up failed.

FR03 Log in

```
Welcome to the Library E-Book Lending & Reservation System!

=== Welcome ===
1. Login
2. Register
3. Exit
Enter your choice: 1

=== Login ===
Enter username: James
Enter password: james123
Login successful.

Welcome, James!

[Logged in as: James | Role: member]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Logout
Enter your choice: █
```

Member log in successfully.

```
Welcome to the Library E-Book Lending & Reservation System!

=== Welcome ===
1. Login
2. Register
3. Exit
Enter your choice: 1

=== Login ===
Enter username: John
Enter password: j
Login successful.

Welcome, john!

[Logged in as: john | Role: admin]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Reservations
5. Users
6. Logout
Enter your choice: █
```

Admin log in successfully.

```

Welcome to the Library E-Book Lending & Reservation System!

=== Welcome ===
1. Login
2. Register
3. Exit
Enter your choice: 1

=== Login ===
Enter username: James2
Enter password: james22
User not found.
Login failed. Invalid username or password.

=== Welcome ===
1. Login
2. Register
3. Exit
Enter your choice: █

```

Sign up with non-existing username - login failed.

My Account/User Pages

FR04 View user details

```

Welcome to the Library E-Book Lending & Reservation System!

=== Welcome ===
1. Login
2. Register
3. Exit
Enter your choice: 1

=== Login ===
Enter username: John
Enter password: j
Login successful.

Welcome, john!

[Logged in as: john | Role: admin]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Reservations
5. Users
6. Logout
Enter your choice: 5

```

```

=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 3
Enter username to view: luna_rose

=== User Details ===
Username: luna_rose
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: █

```

Admin view user details with existing username - view user details successfully.

```

Welcome, john!

[Logged in as: john | Role: admin]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Reservations
5. Users
6. Logout
Enter your choice: 5

=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 3
Enter username to view: david
User not found.

```

Admin view user details with non-existing username - view user details failed.

FR05 Update username

```

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Logout
Enter your choice: 1

=== My Account ===
Username: James
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: 1
Enter new username: JamesLee
Username updated successfully.

=== My Account ===
Username: JamesLee
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: █

```

Member update username with non-existing username - update username successfully.

```
=== My Account ===  
Username: JamesLee  
Role: Member  
1. Update My Username  
2. Update My Password  
3. Delete My Account  
4. Back  
Enter your choice: 1  
Enter new username: JamesLee  
Username update failed: Username already exists.  
  
=== My Account ===  
Username: JamesLee  
Role: Member  
1. Update My Username  
2. Update My Password  
3. Delete My Account  
4. Back  
Enter your choice: █
```

Member update username with existing username - update username failed.

```
=== My Account ===  
Username: JamesLee  
Role: Member  
1. Update My Username  
2. Update My Password  
3. Delete My Account  
4. Back  
Enter your choice: 1  
Enter new username:  
Username update failed: Username cannot be empty.
```

Member update username with empty name - update username failed.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 3
Enter username to view: luna_rose

=== User Details ===
Username: luna_rose
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 1
Enter new username: luna
Username updated successfully.

=== User Details ===
Username: luna
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: █
```

Admin update member/others admin's username with non-existing username - update username successfully.

```
=== User Details ===
Username: luna
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 1
Enter new username: luna
Username update failed: Username already exists.
```

Admin update member/others admin's username with existing username - update username failed.

FR06 Update password

```
=== My Account ===
Username: JamesLee
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: 2
Enter old password: (press [u] to unhide or just press enter to hide): u
Enter old password: : james123
Enter new password: (press [u] to unhide or just press enter to hide): u
Enter new password: : james
Confirm new password: (press [u] to unhide or just press enter to hide): u
Confirm new password: : james
Password updated successfully.
```

Member update password successfully.

```
=== My Account ===
Username: JamesLee
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: 2
Enter old password: (press [u] to unhide or just press enter to hide): u
Enter old password: : james123
Enter new password: (press [u] to unhide or just press enter to hide): u
Enter new password: : james
Confirm new password: (press [u] to unhide or just press enter to hide): u
Confirm new password: : james
Password update failed. Your old password might be wrong.
```

Member update password with wrong old password - update password failed.

```
=== My Account ===
Username: JamesLee
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: 2
Enter old password: (press [u] to unhide or just press enter to hide): u
Enter old password: : james
Enter new password: (press [u] to unhide or just press enter to hide): u
Enter new password: : jamesnewpassword
Confirm new password: (press [u] to unhide or just press enter to hide): u
Confirm new password: : jamespassword
Password update failed. New password and confirmation do not match.
```

Member update password with wrong confirmation password - update password failed.


```
=== User Details ===
Username: luna
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 2
Enter old password: (press [u] to unhide or just press enter to hide): u
Enter old password: : rose123
Enter new password: (press [u] to unhide or just press enter to hide): u
Enter new password: : roseupdate123
Confirm new password: (press [u] to unhide or just press enter to hide): u
Confirm new password: : roseupdate123
Password updated successfully.
```

Admin update member/others admin's password successfully.

```
=== User Details ===
Username: luna
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 2
Enter old password: (press [u] to unhide or just press enter to hide): u
Enter old password: : luna
Enter new password: (press [u] to unhide or just press enter to hide): u
Enter new password: : lunaupdate123
Confirm new password: (press [u] to unhide or just press enter to hide): u
Confirm new password: : lunaupdate123
Password update failed. Your old password might be wrong.
```

Admin update member/others admin's password with wrong old password - update password failed.

```
=== User Details ===
Username: luna
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 2
Enter old password: (press [u] to unhide or just press enter to hide): u
Enter old password: : luna123
Enter new password: (press [u] to unhide or just press enter to hide): u
Enter new password: : lunaupdate123
Confirm new password: (press [u] to unhide or just press enter to hide): u
Confirm new password: : luna123
Password update failed. New password and confirmation do not match.
```

Admin update member/others admin's password with wrong confirmation password - update password failed.

FR07 Update role

```
=== User Details ===
Username: luna
Role: Member
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 3
Enter new role (ADMIN/MEMBER): admin
Role update successfully from 'member' to 'admin'.

=== User Details ===
Username: luna
Role: Admin
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: █
```

Admin update role successfully.

```
=== User Details ===
Username: luna
Role: Admin
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 3
Enter new role (ADMIN/MEMBER): admin
Role update failed. The user already is admin
```

Admin update role with existing role - update role failed.

```
=== User Details ===
Username: luna
Role: Admin
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 3
Enter new role (ADMIN/MEMBER): m
Role update failed. Invalid role entered.
```

Admin update role with invalid input - update role failed.

FR08 Delete user

```
=== My Account ===
Username: JamesLee
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: 3
Are you sure you want to delete your account? (yes/no): yes
Account deleted successfully.
You have been logged out.
```

Member delete account successfully.

```
=== My Account ===
Username: JamesLee
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: 3
Are you sure you want to delete your account? (yes/no): y
Account deletion cancelled.

=== My Account ===
Username: JamesLee
Role: Member
1. Update My Username
2. Update My Password
3. Delete My Account
4. Back
Enter your choice: █
```

Member delete account with invalid input - delete account failed.

```
=== User Details ===
Username: luna
Role: Admin
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: 4
User deleted successfully.

=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: █
```

Admin delete member's account successfully.

FR09 Logout

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 5
Returning to admin dashboard...

[Logged in as: john | Role: admin]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Reservations
5. Users
6. Logout
Enter your choice: 6
You have been logged out.
```

Admin logout successfully.

```
Please select an option:
1. My Account
2. My Reservations
3. Books
4. Logout
Enter your choice: 4
You have been logged out.
```

Member logout successfully.

Reservations Page

FR10 View reservations with pagination

```
= RESERVATIONS - PAGE 1 OF 4 =
```

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
1	js	Ulysses by James Joyce	EXPIRED	2025-08-26T00:00	2025-08-27	2025-09-02
2	luna_rose	Romeo and Juliet by W...	DENIED	2025-08-30T02:19	2025-09-01	2025-11-01
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
4	clark_kent	The lesser Key of Sol...	ACTIVE	2025-08-30T23:27	2025-08-31	2025-09-09
5	clark_kent	The Strange Case of D...	DENIED	2025-08-30T23:27	2025-09-01	2025-10-10

```
[1] Previous | [2] Next | [0] Back
Enter your choice: 2
```

```
= RESERVATIONS - PAGE 2 OF 4 =
```

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
6	bruce_wayne	Narrative of the Life...	DENIED	2025-08-31T00:40	2025-10-11	2025-11-01
7	bruce_wayne	Romeo and Juliet by W...	DENIED	2025-08-31T00:40	2025-09-01	2025-10-10
8	chris_wong	Romeo and Juliet by W...	DENIED	2025-08-31T00:42	2025-09-01	2025-10-01
9	chris_wong	The Wars of Religion ...	DENIED	2025-08-31T00:45	2025-09-01	2025-10-11
10	emma_lee	Meditations by Empero...	DENIED	2025-08-31T00:46	2025-09-01	2025-09-10

```
[1] Previous | [2] Next | [0] Back
Enter your choice: █
```

Admin can view all reservations with pagination

```
= RESERVATIONS - PAGE 1 OF 4 =
```

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
1	js	Ulysses by James Joyce	EXPIRED	2025-08-26T00:00	2025-08-27	2025-09-02
2	luna_rose	Romeo and Juliet by W...	DENIED	2025-08-30T02:19	2025-09-01	2025-11-01
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
4	clark_kent	The lesser Key of Sol...	ACTIVE	2025-08-30T23:27	2025-08-31	2025-09-09
5	clark_kent	The Strange Case of D...	DENIED	2025-08-30T23:27	2025-09-01	2025-10-10

```
[1] Previous | [2] Next | [0] Back
Enter your choice: 7
Invalid choice.
```

```
= RESERVATIONS - PAGE 1 OF 4 =
```

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
1	js	Ulysses by James Joyce	EXPIRED	2025-08-26T00:00	2025-08-27	2025-09-02
2	luna_rose	Romeo and Juliet by W...	DENIED	2025-08-30T02:19	2025-09-01	2025-11-01
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
4	clark_kent	The lesser Key of Sol...	ACTIVE	2025-08-30T23:27	2025-08-31	2025-09-09
5	clark_kent	The Strange Case of D...	DENIED	2025-08-30T23:27	2025-09-01	2025-10-10

```
[1] Previous | [2] Next | [0] Back
Enter your choice: █
```

Admin/ member use invalid page number will show “invalid choice”, it will request them to enter a valid number again.

FR11 Sort reservations

```
--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate): status
Sort descending? (y/n): y

= RESERVATIONS - PAGE 1 OF 4 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 13 | amy_smith | More Letters of Charl... | PENDING | 2025-08-31T00:49 | 2025-09-15 | 2025-10-15 |
| 14 | mike_davis | A Room with a View by... | PENDING | 2025-08-31T00:50 | 2025-09-20 | 2025-10-10 |
| 1 | js | Ulysses by James Joyce | EXPIRED | 2025-08-26T00:00 | 2025-08-27 | 2025-09-02 |
| 2 | luna_rose | Romeo and Juliet by W... | DENIED | 2025-08-30T02:19 | 2025-09-01 | 2025-11-01 |
| 3 | peter_parker | Anne of Green Gables ... | DENIED | 2025-08-30T02:25 | 2025-09-01 | 2025-09-25 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 2

= RESERVATIONS - PAGE 2 OF 4 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 5 | clark_kent | The Strange Case of D... | DENIED | 2025-08-30T23:27 | 2025-09-01 | 2025-10-10 |
| 6 | bruce_wayne | Narrative of the Life... | DENIED | 2025-08-31T00:40 | 2025-10-11 | 2025-11-01 |
| 7 | bruce_wayne | Romeo and Juliet by W... | DENIED | 2025-08-31T00:40 | 2025-09-01 | 2025-10-10 |
| 8 | chris_wong | Romeo and Juliet by W... | DENIED | 2025-08-31T00:42 | 2025-09-01 | 2025-10-01 |
| 9 | chris_wong | The Wars of Religion ... | DENIED | 2025-08-31T00:45 | 2025-09-01 | 2025-10-11 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 2
```

Sort the reservation page by sort status with descending order.

```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate): id
Sort descending? (y/n): n

= RESERVATIONS - PAGE 1 OF 4 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 1 | js | Ulysses by James Joyce | EXPIRED | 2025-08-26T00:00 | 2025-08-27 | 2025-09-02 |
| 2 | luna_rose | Romeo and Juliet by W... | DENIED | 2025-08-30T02:19 | 2025-09-01 | 2025-11-01 |
| 3 | peter_parker | Anne of Green Gables ... | DENIED | 2025-08-30T02:25 | 2025-09-01 | 2025-09-25 |
| 4 | clark_kent | The lesser Key of Sol... | ACTIVE | 2025-08-30T23:27 | 2025-08-31 | 2025-09-09 |
| 5 | clark_kent | The Strange Case of D... | DENIED | 2025-08-30T23:27 | 2025-09-01 | 2025-10-10 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 2

= RESERVATIONS - PAGE 2 OF 4 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 6 | bruce_wayne | Narrative of the Life... | DENIED | 2025-08-31T00:40 | 2025-10-11 | 2025-11-01 |
| 7 | bruce_wayne | Romeo and Juliet by W... | DENIED | 2025-08-31T00:40 | 2025-09-01 | 2025-10-10 |
| 8 | chris_wong | Romeo and Juliet by W... | DENIED | 2025-08-31T00:42 | 2025-09-01 | 2025-10-01 |
| 9 | chris_wong | The Wars of Religion ... | DENIED | 2025-08-31T00:45 | 2025-09-01 | 2025-10-11 |
| 10 | emma_lee | Meditations by Empero... | DENIED | 2025-08-31T00:46 | 2025-09-01 | 2025-09-10 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 

```

Sort the reservation pages by id without descending order.

```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate): username
Sort descending? (y/n): y

      = RESERVATIONS - PAGE 1 OF 4 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 3 | peter_parker | Anne of Green Gables ... | DENIED | 2025-08-30T02:25 | 2025-09-01 | 2025-09-25 |
| 14 | mike_davis | A Room with a View by... | PENDING | 2025-08-31T00:50 | 2025-09-20 | 2025-10-10 |
| 2 | luna_rose | Romeo and Juliet by W... | DENIED | 2025-08-30T02:19 | 2025-09-01 | 2025-11-01 |
| 1 | js | Ulysses by James Joyce | EXPIRED | 2025-08-26T00:00 | 2025-08-27 | 2025-09-02 |
| 16 | js | The Scarlet Letter by... | DENIED | 2025-08-31T21:43 | 2025-09-01 | 2025-10-20 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 2

      = RESERVATIONS - PAGE 2 OF 4 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 17 | js | The Works of Edgar Al... | DENIED | 2025-09-01T15:04 | 2025-09-01 | 2025-10-12 |
| 18 | js | The Enchanted April b... | DENIED | 2025-09-01T16:04 | 2025-09-01 | 2025-10-01 |
| 19 | js | The wonders of scienc... | DENIED | 2025-09-01T16:32 | 2025-09-02 | 2025-10-02 |
| 10 | emma_lee | Meditations by Empero... | DENIED | 2025-08-31T00:46 | 2025-09-01 | 2025-09-10 |
| 11 | emma_lee | The Adventures of Tom... | DENIED | 2025-08-31T00:47 | 2025-09-01 | 2025-11-02 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: █

```

Sort the reservation page by username with descending order.


```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate): booktitle
Sort descending? (y/n): n

```

= RESERVATIONS - PAGE 1 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
14	mike_davis	A Room with a View by...	PENDING	2025-08-31T00:50	2025-09-20	2025-10-10
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
10	emma_lee	Meditations by Empero...	DENIED	2025-08-31T00:46	2025-09-01	2025-09-10
13	amy_smith	More Letters of Charl...	PENDING	2025-08-31T00:49	2025-09-15	2025-10-15
6	bruce_wayne	Narrative of the Life...	DENIED	2025-08-31T00:40	2025-10-11	2025-11-01

[1] Previous | [2] Next | [0] Back
Enter your choice: 2

= RESERVATIONS - PAGE 2 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
2	luna_rose	Romeo and Juliet by W...	DENIED	2025-08-30T02:19	2025-09-01	2025-11-01
7	bruce_wayne	Romeo and Juliet by W...	DENIED	2025-08-31T00:40	2025-09-01	2025-10-10
8	chris_wong	Romeo and Juliet by W...	DENIED	2025-08-31T00:42	2025-09-01	2025-10-01
11	emma_lee	The Adventures of Tom...	DENIED	2025-08-31T00:47	2025-09-01	2025-11-02
18	js	The Enchanted April b...	DENIED	2025-09-01T16:04	2025-09-01	2025-10-01

[1] Previous | [2] Next | [0] Back
Enter your choice: █

Sort the reservation page by book title without descending order.

--- Enter Sort & Filter Options ---

Filter by ID (partial match or leave blank):

Filter by Username (partial match or leave blank):

Filter by Book Title (partial match or leave blank):

Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):

Start Reservation Date (yyyy-mm-dd or leave blank):

End Reservation Date (yyyy-mm-dd or leave blank):

Start Start Date (yyyy-mm-dd or leave blank):

End Start Date (yyyy-mm-dd or leave blank):

Start End Date (yyyy-mm-dd or leave blank):

End End Date (yyyy-mm-dd or leave blank):

Sort by (id, username, booktitle, status, resdate, startdate, enddate): resdate

Sort descending? (y/n): y

= RESERVATIONS - PAGE 1 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
19	js	The wonders of scienc...	DENIED	2025-09-01T16:32	2025-09-02	2025-10-02
18	js	The Enchanted April b...	DENIED	2025-09-01T16:04	2025-09-01	2025-10-01
17	js	The Works of Edgar Al...	DENIED	2025-09-01T15:04	2025-09-01	2025-10-12
16	js	The Scarlet Letter by...	DENIED	2025-08-31T21:43	2025-09-01	2025-10-20
14	mike_davis	A Room with a View by...	PENDING	2025-08-31T00:50	2025-09-20	2025-10-10

[1] Previous | [2] Next | [0] Back

Enter your choice: 2

= RESERVATIONS - PAGE 2 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
13	amy_smith	More Letters of Charl...	PENDING	2025-08-31T00:49	2025-09-15	2025-10-15
11	emma_lee	The Adventures of Tom...	DENIED	2025-08-31T00:47	2025-09-01	2025-11-02
10	emma_lee	Meditations by Empero...	DENIED	2025-08-31T00:46	2025-09-01	2025-09-10
9	chris_wong	The Wars of Religion ...	DENIED	2025-08-31T00:45	2025-09-01	2025-10-11
8	chris_wong	Romeo and Juliet by W...	DENIED	2025-08-31T00:42	2025-09-01	2025-10-01

[1] Previous | [2] Next | [0] Back

Enter your choice:

Sort the reservation page by reservation date with descending order.

```
--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate): startdate
Sort descending? (y/n): y
```

= RESERVATIONS - PAGE 1 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
6	bruce_wayne	Narrative of the Life...	DENIED	2025-08-31T00:40	2025-10-11	2025-11-01
14	mike_davis	A Room with a View by...	PENDING	2025-08-31T00:50	2025-09-20	2025-10-10
13	amy_smith	More Letters of Charl...	PENDING	2025-08-31T00:49	2025-09-15	2025-10-15
19	js	The wonders of scienc...	DENIED	2025-09-01T16:32	2025-09-02	2025-10-02
2	luna_rose	Romeo and Juliet by W...	DENIED	2025-08-30T02:19	2025-09-01	2025-11-01

[1] Previous | [2] Next | [0] Back

Enter your choice: 2

= RESERVATIONS - PAGE 2 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
5	clark_kent	The Strange Case of D...	DENIED	2025-08-30T23:27	2025-09-01	2025-10-10
7	bruce_wayne	Romeo and Juliet by W...	DENIED	2025-08-31T00:40	2025-09-01	2025-10-10
8	chris_wong	Romeo and Juliet by W...	DENIED	2025-08-31T00:42	2025-09-01	2025-10-01
9	chris_wong	The Wars of Religion ...	DENIED	2025-08-31T00:45	2025-09-01	2025-10-11

[1] Previous | [2] Next | [0] Back

Enter your choice: █

Sort the reservation page by start date with descending order.

--- Enter Sort & Filter Options ---

Filter by ID (partial match or leave blank):

Filter by Username (partial match or leave blank):

Filter by Book Title (partial match or leave blank):

Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):

Start Reservation Date (yyyy-mm-dd or leave blank):

End Reservation Date (yyyy-mm-dd or leave blank):

Start Start Date (yyyy-mm-dd or leave blank):

End Start Date (yyyy-mm-dd or leave blank):

Start End Date (yyyy-mm-dd or leave blank):

End End Date (yyyy-mm-dd or leave blank):

Sort by (id, username, booktitle, status, resdate, startdate, enddate): enddate

Sort descending? (y/n): n

= RESERVATIONS - PAGE 1 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
1	js	Ulysses by James Joyce	EXPIRED	2025-08-26T00:00	2025-08-27	2025-09-02
4	clark_kent	The lesser Key of Sol...	ACTIVE	2025-08-30T23:27	2025-08-31	2025-09-09
10	emma_lee	Meditations by Empero...	DENIED	2025-08-31T00:46	2025-09-01	2025-09-10
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
8	chris_wong	Romeo and Juliet by W...	DENIED	2025-08-31T00:42	2025-09-01	2025-10-01

[1] Previous | [2] Next | [0] Back

Enter your choice: 2

= RESERVATIONS - PAGE 2 OF 4 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
18	js	The Enchanted April b...	DENIED	2025-09-01T16:04	2025-09-01	2025-10-01
19	js	The wonders of scienc...	DENIED	2025-09-01T16:32	2025-09-02	2025-10-02
5	clark_kent	The Strange Case of D...	DENIED	2025-08-30T23:27	2025-09-01	2025-10-10
7	bruce_wayne	Romeo and Juliet by W...	DENIED	2025-08-31T00:40	2025-09-01	2025-10-10
14	mike_davis	A Room with a View by...	PENDING	2025-08-31T00:50	2025-09-20	2025-10-10

[1] Previous | [2] Next | [0] Back

Enter your choice:

Sort the reservation page by end date without descending order.

FR12 Filter reservations

```
--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank): 1
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): y
```

= RESERVATIONS - PAGE 1 OF 2 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
19	js	The wonders of scienc...	DENIED	2025-09-01T16:32	2025-09-02	2025-10-02
18	js	The Enchanted April b...	DENIED	2025-09-01T16:04	2025-09-01	2025-10-01
17	js	The Works of Edgar Al...	DENIED	2025-09-01T15:04	2025-09-01	2025-10-12
16	js	The Scarlet Letter by...	DENIED	2025-08-31T21:43	2025-09-01	2025-10-20
14	mike_davis	A Room with a View by...	PENDING	2025-08-31T00:50	2025-09-20	2025-10-10

[1] Previous | [2] Next | [0] Back

Enter your choice: █

Filter reservations by using reservation ID use partial match and descending order.

```
--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank): 1
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): n
```

= RESERVATIONS - PAGE 1 OF 2 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
1	js	Ulysses by James Joyce	EXPIRED	2025-08-26T00:00	2025-08-27	2025-09-02
10	emma_lee	Meditations by Empero...	DENIED	2025-08-31T00:46	2025-09-01	2025-09-10
11	emma_lee	The Adventures of Tom...	DENIED	2025-08-31T00:47	2025-09-01	2025-11-02
13	amy_smith	More Letters of Charl...	PENDING	2025-08-31T00:49	2025-09-15	2025-10-15
14	mike_davis	A Room with a View by...	PENDING	2025-08-31T00:50	2025-09-20	2025-10-10

[1] Previous | [2] Next | [0] Back

Enter your choice: █

Filter reservations by using reservation ID use partial match and without descending order.

```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank): js
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate): username
Sort descending? (y/n): n

      = RESERVATIONS - PAGE 1 OF 1 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 1 | js | Ulysses by James Joyce | EXPIRED | 2025-08-26T00:00 | 2025-08-27 | 2025-09-02 |
| 16 | js | The Scarlet Letter by... | DENIED | 2025-08-31T21:43 | 2025-09-01 | 2025-10-20 |
| 17 | js | The Works of Edgar Al... | DENIED | 2025-09-01T15:04 | 2025-09-01 | 2025-10-12 |
| 18 | js | The Enchanted April b... | DENIED | 2025-09-01T16:04 | 2025-09-01 | 2025-10-01 |
| 19 | js | The wonders of scienc... | DENIED | 2025-09-01T16:32 | 2025-09-02 | 2025-10-02 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 0

=== Reservations Menu ===
1. View All Reservations
2. Sort & Filter Reservations
3. Select Reservation Detail
4. Reserve a Book
0. Back
Enter your choice: 2

```

Filter the reservation page by username with descending order.

```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank): js
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): y

      = RESERVATIONS - PAGE 1 OF 1 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 19 | js | The wonders of scienc... | DENIED | 2025-09-01T16:32 | 2025-09-02 | 2025-10-02 |
| 18 | js | The Enchanted April b... | DENIED | 2025-09-01T16:04 | 2025-09-01 | 2025-10-01 |
| 17 | js | The Works of Edgar Al... | DENIED | 2025-09-01T15:04 | 2025-09-01 | 2025-10-12 |
| 16 | js | The Scarlet Letter by... | DENIED | 2025-08-31T21:43 | 2025-09-01 | 2025-10-20 |
| 1 | js | Ulysses by James Joyce | EXPIRED | 2025-08-26T00:00 | 2025-08-27 | 2025-09-02 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 

```

Filter the reservation page by username without descending order.

```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank): Romeo
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): y

      = RESERVATIONS - PAGE 1 OF 1 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 8 | chris_wong | Romeo and Juliet by W... | DENIED | 2025-08-31T00:42 | 2025-09-01 | 2025-10-01 |
| 7 | bruce_wayne | Romeo and Juliet by W... | DENIED | 2025-08-31T00:40 | 2025-09-01 | 2025-10-10 |
| 2 | luna_rose | Romeo and Juliet by W... | DENIED | 2025-08-30T02:19 | 2025-09-01 | 2025-11-01 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 

```

Filter the reservation page by partially matching the book title without descending order.

```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank): pending
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): y

      = RESERVATIONS - PAGE 1 OF 1 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 14 | mike_davis | A Room with a View by... | PENDING | 2025-08-31T00:50 | 2025-09-20 | 2025-10-10 |
| 13 | amy_smith | More Letters of Charl... | PENDING | 2025-08-31T00:49 | 2025-09-15 | 2025-10-15 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 

```

Filter the reservation page by status without descending order.


```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank): 2025-08-26
End Reservation Date (yyyy-mm-dd or leave blank): 2025-08-31
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n):

```

= RESERVATIONS - PAGE 1 OF 3 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
1	js	Ulysses by James Joyce	EXPIRED	2025-08-26T00:00	2025-08-27	2025-09-02
2	luna_rose	Romeo and Juliet by W...	DENIED	2025-08-30T02:19	2025-09-01	2025-11-01
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
4	clark_kent	The lesser Key of Sol...	ACTIVE	2025-08-30T23:27	2025-08-31	2025-09-09
5	clark_kent	The Strange Case of D...	DENIED	2025-08-30T23:27	2025-09-01	2025-10-10

[1] Previous | [2] Next | [0] Back
Enter your choice: █

Filter the reservation page by reservation date in a rage without descending order.

```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank): 2025-08-31
End Start Date (yyyy-mm-dd or leave blank): 2025-09-10
Start End Date (yyyy-mm-dd or leave blank):
End End Date (yyyy-mm-dd or leave blank):
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): n

```

= RESERVATIONS - PAGE 1 OF 3 =

ID	Username	Book Title	Status	Res. Date	Start Date	End Date
2	luna_rose	Romeo and Juliet by W...	DENIED	2025-08-30T02:19	2025-09-01	2025-11-01
3	peter_parker	Anne of Green Gables ...	DENIED	2025-08-30T02:25	2025-09-01	2025-09-25
4	clark_kent	The lesser Key of Sol...	ACTIVE	2025-08-30T23:27	2025-08-31	2025-09-09
5	clark_kent	The Strange Case of D...	DENIED	2025-08-30T23:27	2025-09-01	2025-10-10
7	bruce_wayne	Romeo and Juliet by W...	DENIED	2025-08-31T00:40	2025-09-01	2025-10-10

[1] Previous | [2] Next | [0] Back
Enter your choice: █

Filter the reservation page by start date in a rage without descending order.


```

--- Enter Sort & Filter Options ---
Filter by ID (partial match or leave blank):
Filter by Username (partial match or leave blank):
Filter by Book Title (partial match or leave blank):
Filter by Status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED or leave blank):
Start Reservation Date (yyyy-mm-dd or leave blank):
End Reservation Date (yyyy-mm-dd or leave blank):
Start Start Date (yyyy-mm-dd or leave blank):
End Start Date (yyyy-mm-dd or leave blank):
Start End Date (yyyy-mm-dd or leave blank): 2025-10-10
End End Date (yyyy-mm-dd or leave blank): 2025-10-15
Sort by (id, username, booktitle, status, resdate, startdate, enddate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): y

= RESERVATIONS - PAGE 1 OF 2 =
-----
| ID | Username | Book Title | Status | Res. Date | Start Date | End Date |
-----
| 17 | js | The Works of Edgar Al... | DENIED | 2025-09-01T15:04 | 2025-09-01 | 2025-10-12 |
| 14 | mike_davis | A Room with a View by... | PENDING | 2025-08-31T00:50 | 2025-09-20 | 2025-10-10 |
| 13 | amy_smith | More Letters of Charl... | PENDING | 2025-08-31T00:49 | 2025-09-15 | 2025-10-15 |
| 9 | chris_wong | The Wars of Religion ... | DENIED | 2025-08-31T00:45 | 2025-09-01 | 2025-10-11 |
| 7 | bruce_wayne | Romeo and Juliet by W... | DENIED | 2025-08-31T00:40 | 2025-09-01 | 2025-10-10 |
-----
[1] Previous | [2] Next | [0] Back
Enter your choice: 

```

Filter the reservation page by end date in a range with descending order.

FR13 Select reservation

```
=== Reservations Menu ===
1. View All Reservations
2. Sort & Filter Reservations
3. Select Reservation Detail
4. Reserve a Book
0. Back
Enter your choice: 3
Enter reservation ID: 11

=== Reservation Detail ===
ID: 11
Username: emma_lee
Book Title: The Adventures of Tom Sawyer, Complete by Mark Twain
Status: DENIED
Reservation Date: 2025-08-31T00:47
Start Date: 2025-09-01
End Date: 2025-11-02

--- Reservation Actions ---
1. View linked book details
2. Update status
3. Update start date
4. Update end date
5. Delete reservation
0. Back
```

Admin can select all reservations.

```
=== Reservations Menu ===
1. View All Reservations
2. Sort & Filter Reservations
3. Select Reservation Detail
4. Reserve a Book
0. Back
Enter your choice: 3
Enter reservation ID: 1

=== Reservation Detail ===
ID: 1
Username: js
Book Title: Ulysses by James Joyce
Status: EXPIRED
Reservation Date: 2025-08-26T00:00
Start Date: 2025-08-27
End Date: 2025-09-02

--- Reservation Actions ---
1. View linked book details
2. Update start date
3. Update end date
0. Back
Your choice: █
```

Users can select their own reservation.

```

Welcome, js!

[Logged in as: js | Role: member]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Logout
Enter your choice: 2

=== Reservations Menu ===
1. View All Reservations
2. Sort & Filter Reservations
3. Select Reservation Detail
4. Reserve a Book
0. Back
Enter your choice: 3
Enter reservation ID: 11
Reservation not found or not accessible.

```

Users can only select their own reservation records. If the entered reservation record does not belong to the user, the message “reservation not found or not accessible” will display

Reservation Page

FR14 View reservation

```

[Logged in as: admin | Role: admin]

Please select an option:
1. My Account
2. My Reservations
3. Books
4. Reservations
5. Users
6. Logout
Enter your choice: 4

=== Reservations Menu ===
1. View All Reservations
2. Sort & Filter Reservations
3. Select Reservation Detail
4. Reserve a Book
0. Back
Enter your choice: 3
Enter reservation ID: 10

=== Reservation Detail ===
ID: 10
Username: emma_lee
Book Title: Meditations by Emperor of Rome Marcus Aurelius
Status: DENIED
Reservation Date: 2025-08-31T00:46
Start Date: 2025-09-01
End Date: 2025-09-10

```

admin/member enable to view the reservations detail

```
=== Reservations Menu ===
1. View All Reservations
2. Sort & Filter Reservations
3. Select Reservation Detail
4. Reserve a Book
0. Back
Enter your choice: 3
Enter reservation ID: 20
Reservation not found or not accessible

=== Reservations Menu ===
1. View All Reservations
2. Sort & Filter Reservations
3. Select Reservation Detail
4. Reserve a Book
0. Back
Enter your choice: |
```

Reservations cannot be found, back to the reservations menu. When the reservations do not exist and the user is not an authorized user (admin) .

FR15 View book

```
--- Reservation Actions ---
1. View linked book details
2. Update start date
3. Update end date
0. Back
Your choice: 1

                                = BOOK DETAILS =

ID: 47
Title: Ulysses by James Joyce
Author: 8 by James Joyce
Genre: Fiction
Released: 2003-07-01
Filename: 4300_Ulysses_by_James_Joyce.txt

Press Enter to go back...

=== Reservation Detail ===
ID: 1
Username: js
Book Title: Ulysses by James Joyce
Status: EXPIRED
Reservation Date: 2025-08-26T00:00
Start Date: 2025-08-27
End Date: 2025-09-02

--- Reservation Actions ---
1. View linked book details
2. Update start date
3. Update end date
0. Back
Your choice: █
```

admin / member can view the book details, and press enter it will direct the user to the "Reservation Detail Page".

FR16 Update status

```
=== Reservation Detail ===
ID: 21
Username: js
Book Title: Leviathan by Thomas Hobbes
Status: PENDING
Reservation Date: 2025-09-05T21:21
Start Date: 2025-09-05
End Date: 2025-09-15

--- Reservation Actions ---
1. View linked book details
2. Update status
3. Update start date
4. Update end date
5. Delete reservation
0. Back
Your choice: 2
Enter new status (ACTIVE/CANCELLED/COMPLETED/PENDING/DENIED): DENIED
Status updated!

=== Reservation Detail ===
ID: 21
Username: js
Book Title: Leviathan by Thomas Hobbes
Status: DENIED
Reservation Date: 2025-09-05T21:21
Start Date: 2025-09-05
End Date: 2025-09-15
```

Admin update the reservation status of the user.

FR17 Update start date

```
=== Reservation Detail ===
ID: 22
Username: js
Book Title: My Life ʄÇö Volume 1 by Richard Wagner
Status: PENDING
Reservation Date: 2025-09-05T21:33
Start Date: 2025-09-05
End Date: 2025-09-25

--- Reservation Actions ---
1. View linked book details
2. Update start date
3. Update end date
0. Back
Your choice: 2
Enter new start date (YYYY-MM-DD): 2025-09-06
Start date updated!

=== Reservation Detail ===
ID: 22
Username: js
Book Title: My Life ʄÇö Volume 1 by Richard Wagner
Status: PENDING
Reservation Date: 2025-09-05T21:33
Start Date: 2025-09-06
End Date: 2025-09-25
```

Users/admin can update the start date.

```
--- Reservation Actions ---
1. View linked book details
2. Update start date
3. Update end date
0. Back
Your choice: 2
Enter new start date (YYYY-MM-DD): 2025-10-01
Failed.

=== Reservation Detail ===
ID: 22
Username: js
Book Title: My Life ΓÇö Volume 1 by Richard Wagner
Status: PENDING
Reservation Date: 2025-09-05T21:33
Start Date: 2025-09-06
End Date: 2025-09-15
```

If the start date is later than the end date, “Failed” will be displayed.

FR18 Update end date

```
--- Reservation Actions ---
1. View linked book details
2. Update start date
3. Update end date
0. Back
Your choice: 3
Enter new end date (YYYY-MM-DD): 2025-09-15
End date updated!

=== Reservation Detail ===
ID: 22
Username: js
Book Title: My Life ΓÇö Volume 1 by Richard Wagner
Status: PENDING
Reservation Date: 2025-09-05T21:33
Start Date: 2025-09-06
End Date: 2025-09-15
```

Users/admin can update the end date.

```

--- Reservation Actions ---
1. View linked book details
2. Update start date
3. Update end date
0. Back
Your choice: 3
Enter new end date (YYYY-MM-DD): 2025-09-01
Failed.

=== Reservation Detail ===
ID: 22
Username: js
Book Title: My Life ʘö Volume 1 by Richard Wagner
Status: PENDING
Reservation Date: 2025-09-05T21:33
Start Date: 2025-09-06
End Date: 2025-09-15

```

If the end date is earlier than the start date, “Failed” will be displayed.

FR19 Delete reservation

```

=== Reservation Detail ===
ID: 21
Username: js
Book Title: Leviathan by Thomas Hobbes
Status: DENIED
Reservation Date: 2025-09-05T21:21
Start Date: 2025-09-05
End Date: 2025-09-15

--- Reservation Actions ---
1. View linked book details
2. Update status
3. Update start date
4. Update end date
5. Delete reservation
0. Back
Your choice: 5
Deleted successfully!

```

Admin can delete the reservation

Books page

FR20 View books with pagination

= BOOK LIBRARY - PAGE 1 OF 21 =				
ID	Title	Author	Genre	Released
1	Moby Dick; Or, The Whale by Herm...	28 by Herman Melville	Fiction	2001-07-01
2	Frankenstein; Or, The Modern Pro...	30 by Mary Wollstonecr...	Fiction	1993-10-01
3	Romeo and Juliet by William Shak...	336 by William Shakesp...	Fiction	1998-11-01
4	A Room with a View by E. M. Forster	15 by E. M. Forster	Fiction	2001-05-01
5	The Complete Works of William Sh...	336 by William Shakesp...	Fiction	1994-01-01
Navigation:				
2: Next				
0: Back to Books (Home)				
Choose an option: 2				
= BOOK LIBRARY - PAGE 2 OF 21 =				
ID	Title	Author	Genre	Released
6	How to Observe_Morals and Manne...	29 by Harriet Martineau	Fiction	2010-10-05
7	Pride and Prejudice by Jane Austen	33 by Jane Austen	Fiction	1998-06-01
8	Middlemarch by George Eliot	30 by George Eliot	Fiction	1994-07-01
9	Little Women; Or, Meg, Jo, Beth,...	62 by Louisa May Alcott	Fiction	2011-08-16
10	Beowulf_ An Anglo-Saxon Epic Poe...	Hall, J. Lesslie (John...	Fiction	2005-07-19
Navigation:				
1: Previous				
2: Next				
0: Back to Books (Home)				
Choose an option: 2				

View Books by Pagination, 5 Books a Page

FR21 Sort books

```
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
5. Back to Main Menu
Enter your choice: 2
Filter by ID (partial match): 1
Filter by Title (partial match):
Filter by Author (partial match):
Filter by Genre (partial match):
Start Release Date (yyyy-mm-dd or blank):
End Release Date (yyyy-mm-dd or blank):
Sort by (id, title, author, genre, releasedDate): id
Sort descending? (y/n):

= BOOK LIBRARY - PAGE 1 OF 5 =
```

ID	Title	Author	Genre	Released
02	Test Book	Tharmu	Fiction	2005-09-19
01	Hi	Hi	Fantasy	2923-12-21
00	The Romance of Lust_ A classic V...	936 by Anonymous	Fiction	2009-10-14
01	Little Women by Louisa May Alcott	62 by Louisa May Alcott	Fiction	1996-05-01
01	The Wonderful Wizard of Oz by L....	115 by L. Frank Baum	Fiction	1993-02-01

```
Navigation:
2: Next
0: Back to Books (Home)
Choose an option: 
```

Sort books by ID Partial Match

```
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
5. Back to Main Menu
Enter your choice: 2
Filter by ID (partial match):
Filter by Title (partial match): The
Filter by Author (partial match):
Filter by Genre (partial match):
Start Release Date (yyyy-mm-dd or blank):
End Release Date (yyyy-mm-dd or blank):
Sort by (id, title, author, genre, releasedDate):
Invalid sort field. Using 'id' as default.
Sort descending? (y/n): n

= BOOK LIBRARY - PAGE 1 OF 10 =
```

ID	Title	Author	Genre	Released
	Moby Dick; Or, The Whale by Herm...	28 by Herman Melville	Fiction	2001-07-01
	Frankenstein; Or, The Modern Pro...	30 by Mary Wollstonecr...	Fiction	1993-10-01
	The Complete Works of William Sh...	336 by William Shakesp...	Fiction	1994-01-01
3	The Enchanted April by Elizabeth...	16 by Elizabeth Von Arnim	Fiction	2005-07-29
4	The Blue Castle_ a novel by L. M...	33 by L. M. Montgomery	Fiction	2022-05-03

```
Navigation:
2: Next
0: Back to Books (Home)
Choose an option: 
```

Sort books by Title Partial Match

```
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: 2
Filter by ID (partial match):
Filter by Title (partial match):
Filter by Author (partial match): C
Filter by Genre (partial match):
Start Release Date (yyyy-mm-dd or blank):
End Release Date (yyyy-mm-dd or blank):
Sort by (id, title, author, genre, releasedDate): id
Sort descending? (y/n):

= BOOK LIBRARY - PAGE 1 OF 7 =

-----
ID      | Title                                     | Author                                     | Genre   | Released
-----
99      | Meditations by Emperor of Rome M...    | 7 by Emperor of Rome M... | Fiction | 2001-06-01
97      | Ancient rhetoric and poetic _ In...    | 3 by Charles Sears Bal... | Fiction | 2025-08-19
95      | Shakespeare's family by C. C. St...    | 3 by C. C. Stopes        | Fiction | 2008-08-14
93      | Frankenstein; Or, The Modern Pro...    | 30 by Mary Wollstonecr... | Fiction | 2012-11-23
92      | The History of the Peloponnesian...    | 11 by Thucydides         | Fiction | 2004-12-01
-----

Navigation:
2: Next
0: Back to Books (Home)
Choose an option: 
```

Sort books by Author Partial Match

```
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: 2
Filter by ID (partial match):
Filter by Title (partial match):
Filter by Author (partial match):
Filter by Genre (partial match): Fiction
Start Release Date (yyyy-mm-dd or blank):
End Release Date (yyyy-mm-dd or blank):
Sort by (id, title, author, genre, releasedDate): id
Sort descending? (y/n):

= BOOK LIBRARY - PAGE 1 OF 20 =

-----
ID      | Title                                     | Author                                     | Genre   | Released
-----
102     | Test Book                               | Tharmu                                | Fiction | 2005-09-19
100     | The Romance of Lust_ A classic V...    | 936 by Anonymous            | Fiction | 2009-10-14
99      | Meditations by Emperor of Rome M...    | 7 by Emperor of Rome M... | Fiction | 2001-06-01
98      | Heath's Modern Language Series_ ...    | Nelson, Ernesto             | Fiction | 2012-05-07
97      | Ancient rhetoric and poetic _ In...    | 3 by Charles Sears Bal... | Fiction | 2025-08-19
-----

Navigation:
2: Next
0: Back to Books (Home)
Choose an option: 
```

Sort books by Genre

```

BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: 2
Filter by ID (partial match): 1
Filter by Title (partial match): The
Filter by Author (partial match):
Filter by Genre (partial match): Fiction
Start Release Date (yyyy-mm-dd or blank):
End Release Date (yyyy-mm-dd or blank):
Sort by (id, title, author, genre, releasedDate): id
Sort descending? (y/n): y

= BOOK LIBRARY - PAGE 1 OF 2 =
-----
ID      | Title                                     | Author                               | Genre  | Released
-----
100     | The Romance of Lust_ A classic V...    | 936 by Anonymous                   | Fiction | 2009-10-14
81      | The Wonderful Wizard of Oz by L....    | 115 by L. Frank Baum               | Fiction | 1993-02-01
61      | Wuthering Heights by Emily Bront|    | 3 by Emily Bront|                    | Fiction | 1996-12-01
18      | The Expedition of Humphry Clinker...   | 28 by T. Smollett                  | Fiction | 2000-04-01
17      | The Adventures of Ferdinand Coun...    | 28 by T. Smollett                  | Fiction | 2004-09-09
-----
Navigation:
2: Next
0: Back to Books (Home)

```

Sort the book by id using partial match and with descending order.

FR22 Filter books

```

BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: Enter your choice: 2
Filter by ID (partial match): 3
Filter by Title (partial match): T
Filter by Author (partial match):
Filter by Genre (partial match): Fiction
Start Release Date (yyyy-mm-dd or blank):
End Release Date (yyyy-mm-dd or blank):

```

Filter books by using title, ID and genre using partial match

FR23 Select book

```
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: 3
Enter Book ID to view details (0 to go back): 78

= BOOK DETAILS =
ID: 78
Title: Beyond Good and Evil by Friedrich Wilhelm Nietzsche
Author: 43 by Friedrich Wilhelm Nietzsche
Genre: Fiction
Released: 2003-08-01
Filename: 4363_Beyond_Good_and_Evil_by_Friedrich_Wilhelm_Nietzsche.txt
BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: █
```

Select books by ID

FR24 Add book

```
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: 4
Enter title: TestTitlee
Enter author: Alicia
Enter genre: Non-Fiction
Enter release date (yyyy-mm-dd):
Invalid date format. Please use yyyy-mm-dd.
Enter release date (yyyy-mm-dd):
Invalid date format. Please use yyyy-mm-dd.
Enter release date (yyyy-mm-dd): 2005-09-19
Enter filename (without .txt): sample
```

Adding books with error checking

Book page

FR25 View book

```
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: Enter your choice: 3
Enter Book ID to view details (0 to go back): 56

= BOOK DETAILS =

ID: 56
Title: Great Expectations by Charles Dickens
Author: 211 by Charles Dickens
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 5
Invalid choice.
```

Viewing Books Details with Options

```
Enter Book ID to view details (0 to go back): 10101
Book not found.
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
```

Invalid ID Check

```

                                = BOOK DETAILS =
ID: 56
Title: Great Expectations by Charles Dickens
Author: 211 by Charles Dickens
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
      BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 4
  Book 'Great Expectations by Charles Dickens' reserved successfully!

```

Reserve Books that is selected

```
= BOOK DETAILS =
ID: 56
Title: Great Expectations by Charles Dickens
Author: 211 by Charles Dickens
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 3

READING - PAGE 1
The Project Gutenberg eBook of Great Expectations

This ebook is for the use of anyone anywhere in the United States and
most other parts of the world at no cost and with almost no restrictions
whatsoever. You may copy it, give it away or re-use it under the terms
of the Project Gutenberg License included with this ebook or online
at www.gutenberg.org. If you are not located in the United States,
you will have to check the laws of the country where you are located
before using this eBook.

Title: Great Expectations

Author: Charles Dickens

Release date: July 1, 1998 [eBook #1400]
Most recently updated: December 17, 2024

Language: English

Credits: An Anonymous Volunteer and David Widger
Navigation:
2: Next
0: Exit
Option: 2
```


FR28 Update title

```

= BOOK DETAILS =
ID: 56
Title: Great Expectations by Charles Dickens
Author: 211 by Charles Dickens
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 1
Enter field to update (title, author, genre, date, filename): title
Enter new value: Great_Expectations by Charles Dickens
Updated successfully.

= BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: 211 by Charles Dickens
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: █
```

Update Title

FR29 Update author

```

= BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: 211 by Charles Dickens
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
    BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 1
Enter field to update (title, author, genre, date, filename): author
Enter new value: Tharmu
Updated successfully.

= BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
    BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 
```

Updating Author

FR30 Update genre

```

                                = BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
                                BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 1
Enter field to update (title, author, genre, date, filename): genre
Enter new value: Science Fiction
Updated successfully.

                                = BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Science Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
                                BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 
```

Updating Genre

FR31 Update release date

```

                                = BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Science Fiction
Released: 1998-07-01
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
    BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 1
Enter field to update (title, author, genre, date, filename): date
Enter new release date (yyyy-mm-dd): 2005-05-15
    Updated successfully.
```

```

                                = BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Science Fiction
Released: 2005-05-15
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
    BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 
```

Updating Release Date

FR32 Update filename

```

                                = BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Science Fiction
Released: 2005-05-15
Filename: 1400_Great_Expectations_by_Charles_Dickens.txt
      BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 1
Enter field to update (title, author, genre, date, filename): filename
Enter new value: sample
Updated successfully.

                                = BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Science Fiction
Released: 2005-05-15
Filename: sample.txt
      BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: █
```

Update filename

```

= BOOK DETAILS =
ID: 67
Title: Walden, and On The Duty Of Civil Disobedience by Henry David Thoreau
Author: 24 by Henry David Thoreau
Genre: Fiction
Released: 1995-01-01
Filename: 205_Walden,_and_On_The_Duty_Of_Civil_Disobedience_by_Henry_David_Thoreau.txt
BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 1
Enter field to update (title, author, genre, date, filename): filename
Enter new value: 6|
Invalid filename. Use only letters, numbers, _, - and no : ? * < > |
Enter new filename: filenamedoesn't exist

```

Update filename error checking for invalid character

FR33 Delete book

```

= BOOK DETAILS =
ID: 56
Title: Great_Expectations by Charles Dickens
Author: Tharmu
Genre: Science Fiction
Released: 2005-05-15
Filename: sample.txt
BOOK OPTIONS
1. Update Book (Admin)
2. Delete Book (Admin)
3. Read Book
4. Reserve Book
0. Back
Choose an option: 2
Type 'YES' to confirm deletion: YES
Book deleted successfully.
BOOKS (HOME PAGE)
1. View Books by Page
2. Filter & Sort Books
3. View Book Details
4. Create New Book (Admin Only)
0. Back to Main Menu
Enter your choice: Enter your choice: 

```

Users page

FR34 View users with pagination

```
=== Login ===
```

```
Enter username: john
```

```
Enter password: j
```

```
Login successful.
```

```
Welcome, john!
```

```
[Logged in as: john | Role: admin]
```

```
Please select an option:
```

```
1. My Account
```

```
2. My Reservations
```

```
3. Books
```

```
4. Reservations
```

```
5. Users
```

```
6. Logout
```

```
Enter your choice: 5
```

```
=== User Management (Admin) ===
```

```
1. View Users by Page
```

```
2. Filter & Sort Users
```

```
3. View User Details
```

```
4. Register New User
```

```
5. Back
```

```
Enter your choice: 1
```

```
=== User List ===
```

```
Username
```

```
Role
```

```
James
```

```
member
```

```
Sandy
```

```
member
```

```
admin
```

```
admin
```

```
amy_smith
```

```
member
```

```
bruce_wayne
```

```
member
```

```
chris_wong
```

```
member
```

```
clark_kent
```

```
member
```

```
david_clark
```

```
admin
```

```
emma_lee
```

```
member
```

```
john
```

```
admin
```

```
Page (1/2)
```

```
Enter page number (1-2) or 0 to go back: 2
```

```
=== User List ===
```

```
Username
```

```
Role
```

```
js
```

```
member
```

```
mike_davis
```

```
member
```

```
mod_stephen
```

```
admin
```

```
peter_parker
```

```
member
```

```
sara_jones
```

```
member
```

```
techie_ash
```

```
member
```

```
tony_stark
```

```
admin
```

```
Page (2/2)
```

```
Enter page number (1-2) or 0 to go back: 0
```

```
=== User Management (Admin) ===
```

```
1. View Users by Page
```

```
2. Filter & Sort Users
```

```
3. View User Details
```

```
4. Register New User
```

```
5. Back
```

```
Enter your choice: █
```

View users with pagination successfully.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 1

=== User List ===
Username          Role
-----
James             member
Sandy             member
admin             admin
amy_smith         member
bruce_wayne       member
chris_wong        member
clark_kent        member
david_clark       admin
emma_lee          member
john              admin

Page (1/2)
Enter page number (1-2) or 0 to go back: 3
Invalid page number.
```

View users with invalid page number - view users with pagination failed - request to enter page number again.

FR35 Sort users

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 2
Enter username filter (press enter to skip):
Enter role filter (ADMIN/USER, press enter to skip):
Sort by (username/role): username
Order (asc/desc): asc

=== User List ===
Username          Role
-----
James             member
Sandy             member
admin             admin
amy_smith         member
bruce_wayne       member
chris_wong        member
clark_kent        member
david_clark       admin
emma_lee          member
john              admin

Page (1/2)
Enter page number (1-2) or 0 to go back: █
```

Sort users by username with ascending order successfully(default).

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 2
Enter username filter (press enter to skip):
Enter role filter (ADMIN/USER, press enter to skip):
Sort by (username/role): username
Order (asc/desc): desc

=== User List ===
Username          Role
-----
tony_stark         admin
techie_ash         member
sara_jones         member
peter_parker       member
mod_stephen        admin
mike_davis         member
js                member
john              admin
emma_lee          member
david_clark        admin

Page (1/2)
Enter page number (1-2) or 0 to go back: █
```

Sort users by username with descending order successfully.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 2
Enter username filter (press enter to skip):
Enter role filter (ADMIN/USER, press enter to skip):
Sort by (username/role): role
Order (asc/desc): asc

=== User List ===
Username          Role
-----
admin             admin
mod_stephen       admin
david_clark       admin
tony_stark        admin
john              admin
js                member
techie_ash        member
sara_jones        member
mike_davis        member
emma_lee          member

Page (1/2)
Enter page number (1-2) or 0 to go back: █
```

Sort users by role with ascending order successfully.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 2
Enter username filter (press enter to skip):
Enter role filter (ADMIN/USER, press enter to skip):
Sort by (username/role): role
Order (asc/desc): desc

=== User List ===
Username          Role
-----
js                member
techie_ash        member
sara_jones        member
mike_davis        member
emma_lee          member
chris_wong        member
amy_smith         member
peter_parker      member
bruce_wayne       member
clark_kent        member

Page (1/2)
Enter page number (1-2) or 0 to go back: █
```

Sort users by role with descending order successfully.

FR36 Filter user

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 2
Enter username filter (press enter to skip): c
Enter role filter (ADMIN/USER, press enter to skip):
Sort by (username/role):
Order (asc/desc): asc

=== User List ===
Username          Role
-----
bruce_wayne       member
chris_wong        member
clark_kent        member
david_clark       admin
techie_ash        member

Page (1/1)
Enter page number (1-1) or 0 to go back: 
```

Filter by username with ascending order successfully.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 2
Enter username filter (press enter to skip):
Enter role filter (ADMIN/USER, press enter to skip): admin
Sort by (username/role):
Order (asc/desc): desc

=== User List ===
Username          Role
-----
tony_stark        admin
mod_stephen       admin
john              admin
david_clark       admin
admin             admin

Page (1/1)
Enter page number (1-1) or 0 to go back: 
```

Filter by role(admin) with descending order successfully.

FR37 Select user

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 3
Enter username to view: tony_stark

=== User Details ===
Username: tony_stark
Role: Admin
1. Update Username
2. Update User Password
3. Update User Role
4. Delete User Account
5. Back
Enter your choice: █
```

Select user with existing username - select user successfully - view user details.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 3
Enter username to view: tony
User not found.
```

Select user with non-existing username - select user failed.

FR38 Create user

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 4
Enter new username: ron
Enter password: (press [u] to unhide or just press enter to hide): u
Enter password: : weasley
Confirm password: (press [u] to unhide or just press enter to hide): u
Confirm password: : weasley
Enter role (ADMIN/MEMBER): member
User created successfully.
```

Users create successfully.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 4
Enter new username: luna
Enter password: (press [u] to unhide or just press enter to hide):
Enter password: :

Confirm password: (press [u] to unhide or just press enter to hide):
Confirm password: :

Enter role (ADMIN/MEMBER): member
User creation failed. User might exist.
```

Create a user with an existing username - user creation failed.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 4
Enter new username: ron
Enter password: (press [u] to unhide or just press enter to hide): u
Enter password: : weasley
Confirm password: (press [u] to unhide or just press enter to hide): u
Confirm password: : wesley
User creation failed. Password and Confirmation Password do not match.
```

Create a user with a wrong confirmation password - user creation failed.

```
=== User Management (Admin) ===
1. View Users by Page
2. Filter & Sort Users
3. View User Details
4. Register New User
5. Back
Enter your choice: 4
Enter new username: Dumbledore
Enter password: (press [u] to unhide or just press enter to hide):
Enter password: :

Confirm password: (press [u] to unhide or just press enter to hide):
Confirm password: :

Enter role (ADMIN/MEMBER):
Invalid role entered. Please enter either ADMIN or MEMBER.
Enter role (ADMIN/MEMBER): admin
User created successfully.
```

Create a user must enter their role - user create successfully.